

IIR FILTER DESIGN

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IIR Filter : ✓

$$H(s) \xrightarrow[\text{BLT} \checkmark]{\text{Impulse Inv.}} H(z)$$

Impulse Invariance (II)

$$\frac{1}{s - \alpha}$$

$$\frac{1}{1 - e^{\alpha T} z^{-1}}$$

$$\frac{s + \alpha}{(s + \alpha)^2 + \beta^2}$$

$$\frac{1 - e^{-\alpha T} (\cos \beta T) z^{-1}}{1 - 2e^{-\alpha T} (\cos \beta T) z^{-1} + e^{-2\alpha T} z^{-2}}$$

$$\frac{\beta}{(s + \alpha)^2 + \beta^2}$$

$$\frac{e^{-\alpha T} (\sin \beta T) z^{-1}}{1 - 2e^{-\alpha T} (\cos \beta T) z^{-1} + e^{-2\alpha T} z^{-2}}$$

Ex: $H_a(s) = \frac{2}{(s+1)(s+2)}$, Find $H(z)$ using Impulse Invariance.

1st applying partial fraction

$$\frac{A}{(s+1)} + \frac{B}{(s+2)}$$

$$A(s+2) + B(s+1) = 2$$

$$\text{when } s = -1$$

$$A \cdot 1 = 2$$

$$A = 2$$

$$\text{when } s = -2$$

$$B = -2$$

putting values of A & B

$$= \frac{2}{s+1} - \frac{2}{s+2}$$

$$= 2 \cdot \frac{1}{s-(-1)} - 2 \cdot \frac{1}{s-(-2)}$$

$$= 2 \cdot \frac{1}{1 - e^{-1T} z^{-1}} - 2 \cdot \frac{1}{1 - e^{-2T} z^{-1}}$$

$$= 2 \cdot \frac{1}{1 - e^{-1} z^{-1}} - 2 \cdot \frac{1}{1 - e^{-2} z^{-1}}$$

$$H(z) = \frac{2}{1 - 0.36 z^{-1}} - \frac{2}{1 - 0.135 z^{-1}}$$

$H_a(s) = \frac{1}{(s+0.5)(s^2+0.5s+2)}$

⇒ Using partial fraction,

$$\frac{A}{s+0.5} + \frac{Bs+C}{s^2+0.5s+2} \quad \text{--- (i)}$$

$$H_a(s) = \frac{A(s^2+0.5s+2) + (Bs+C)(s+0.5)}{(s+0.5)(s^2+0.5s+2)}$$

$$1 = A(s^2+0.5s+2) + (Bs+C)(s+0.5)$$

$$1 = As^2 + 0.5As + 2A + Bs^2 + 0.5Bs + Cs + 0.5C$$

$$1 = As^2 + Bs^2 + 0.5As + 0.5Bs + Cs + 2A + 0.5C$$

$$1 = (A+B)s^2 + (0.5A+0.5B+C)s + (2A+0.5C)$$

In case, if
Sampling freq is given
like 1 kHz.

$$\text{Then } T = \frac{1}{f_s}$$

$$T = \frac{1}{1000} =$$

$$\text{let } T = 1 \text{ sec.}$$

Now,

$$\begin{aligned} A+B &= 0 & 0.5A+0.5B+C &= 0 & 2A+0.5C &= 1 \\ A &= -B & -0.5B+0.5B+C &= 0 & -2B+0.5C &= 1 \\ & & C &= 0 & -2B+0 &= 1 \\ & & & & B &= -\frac{1}{2} \end{aligned}$$

$$\therefore A = +\frac{1}{2}$$

putting values of A, B & C in eqⁿ (i)

$$\frac{0.5}{s+0.5} - \frac{0.5s}{s^2+0.5s+2}$$

$$= \frac{0.5}{1-e^{-0.5\tau} z^{-1}} - \frac{0.5s}{s^2+0.5s+2} \quad \text{--- (A)}$$

Now. Doing for

$$\frac{0.5s}{s^2+0.5s+2}$$

$$= \frac{0.5s}{s^2 + 2 \cdot s \cdot \frac{1}{4} + \left(\frac{1}{4}\right)^2 - \left(\frac{1}{4}\right)^2 + 2}$$

$$= \frac{0.5s}{\left(s+\frac{1}{4}\right)^2 + 1.93}$$

$$\left[2 - \left(\frac{1}{4}\right)^2\right]$$

$$= \frac{0.5s}{(s+0.25)^2 + (1.39)^2}$$

$$= \frac{0.5[s+0.25-0.25]}{(s+0.25)^2 + (1.39)^2}$$

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$$= \frac{0.5 \cdot \frac{s+0.25}{(s+0.25)^2 + (1.39)^2}}{\frac{0.5 \times 0.25}{(s+0.25)^2 + (1.39)^2}}$$

$$= \frac{0.5 \cdot \frac{1 - e^{-0.25T} \cos(1.39T) z^{-1}}{1 - 2e^{-0.25T} \cos(1.39T) z^{-1} + e^{-0.5T} z^{-2}}}{\frac{0.5 \times 0.25}{1.39} \times \frac{1.39}{(s+0.25)^2 + (1.39)^2}}$$

$$\frac{0.5 \times 0.25}{1.39} \times \frac{1.39}{(s+0.25)^2 + (1.39)^2}$$

$$= \frac{0.5 \cdot \frac{1 - e^{-0.25T} \cos(1.39T) z^{-1}}{1 - 2e^{-0.25T} \cos(1.39T) z^{-1} + e^{-0.5T} z^{-2}}}{\frac{0.5 \times 0.25}{1.39} \times \frac{1.39}{(s+0.25)^2 + (1.39)^2}}$$

$$0.08 \cdot \frac{e^{-0.25T} \sin(1.39T) z^{-1}}{1 - 2e^{-0.5T} \cos(1.39T) z^{-1} + e^{-0.5T} z^{-2}}$$

from eqn - (A)

$$H(z) = \frac{0.5}{1 - e^{-0.5T} z^{-1}} - \frac{0.5 \cdot \frac{1 - e^{-0.25T} \cos(1.39T) z^{-1}}{1 - 2e^{-0.25T} \cos(1.39T) z^{-1} + e^{-0.5T} z^{-2}}}{\frac{0.5 \times 0.25}{1.39} \times \frac{1.39}{(s+0.25)^2 + (1.39)^2}} + \frac{0.08 \cdot \frac{e^{-0.25T} \sin(1.39T) z^{-1}}{1 - 2e^{-0.5T} \cos(1.39T) z^{-1} + e^{-0.5T} z^{-2}}}{\frac{0.5 \times 0.25}{1.39} \times \frac{1.39}{(s+0.25)^2 + (1.39)^2}}$$

let $T=1$

then calculate $H(z)$

$$H(z) =$$

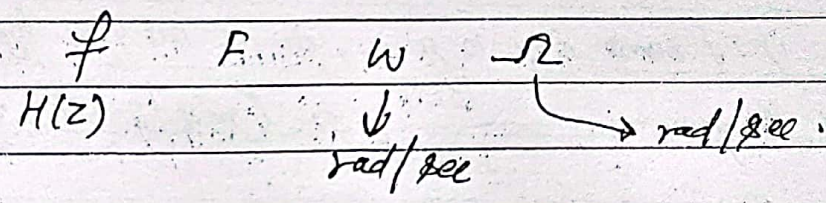
BLT

$H(s) \xrightarrow{\text{BLT}} H(z)$

$$s \rightarrow \frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right]$$

$$H_a(s) = \frac{1}{(s+0.5)(s^2+0.5s+2)}$$

$$H(z) = \frac{1}{\left[\frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 0.5 \right] \left[\left\{ \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right\}^2 + 0.5 \left\{ \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right\} + 2 \right]}$$



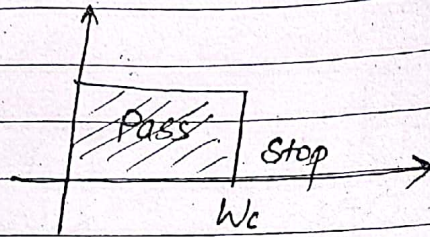
BLT

Impulse Invariance

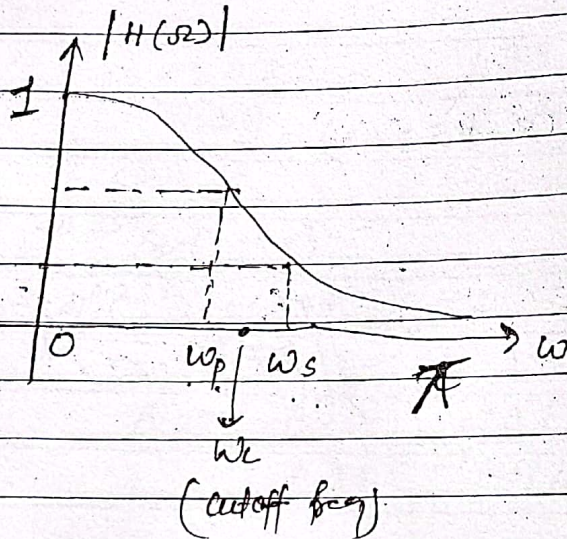
$$\Omega = \frac{2}{T} \tan \frac{\omega}{2}$$

$$\Omega = \frac{\omega}{T}$$

Butterworth Approx:



Ripple & Attenuation are same



Same Pass band edge
same

Steps: 1) If samp. freq is given, divide all freq by samp. freq.

2) Convert all frequencies $\xrightarrow[\text{TL}]{\text{BLT}}$ ω (rad/sec)

Butter Filter Design

1) calculate Order (N)

$$N = \frac{\log \left[\frac{10^{\frac{A_s}{10}} - 1}{10^{\frac{A_p}{10}} - 1} \right]}{2 \log (\omega_s / \omega_p)}$$

If question $(A_p, A_s \geq \text{dB})$

$$N = \frac{\log \left[\frac{1/A_s^2 - 1}{1/A_p^2 - 1} \right]}{2 \log (\omega_s / \omega_p)}$$

if not given in dB, use this formula.

ii) Cut off frequency .

$$\times \quad \Omega_c = \frac{\Omega_p}{\left[\frac{1}{A_p^2} - 1 \right]^{\frac{1}{2N}}} \quad , \text{ if dB not given in dB}$$

$$\times \quad \Omega_c = \frac{\Omega_p}{\left[10^{\frac{A_p}{10}} - 1 \right]^{\frac{1}{2N}}} \quad , \text{ if dB given in dB}$$

iii) Transfer function Poles

$$P_i = + \Omega_c e^{j \frac{(N+2i+1)\pi}{2N}} \quad , \quad i = 0 \dots N-1$$

iv) Transfer function :

$$H(s) = \frac{\Omega_c^N}{(s - p_0)(s - p_1) \dots (s - p_N)}$$

If Sam freq is not given then, f_c F_0
 cutoff freq is (f_c) $\downarrow$
 Discrete Continuous

Design a second order discrete time Butterworth filter with cutoff frequency 1 kHz and sampling frequency of 10^4 sample/sec by using Bilinear Transformation.

⇒ Soln
 Given, Sampling Frequency (F_s) = 10^4 sample/sec.
 cutoff frequency (F_c) = 1 kHz
 = 1000 Hz.

after sampling, we find f_c

$$f_c \cdot \frac{F_c}{F_s} = \frac{10^3}{10^4}$$

$$= 0.1$$

For B.L.T, we have

$$\Omega = \frac{2}{T} \tan \frac{\omega}{2} \quad \text{--- (1)}$$

$$\omega_c = 2\pi f_c$$

$$\omega_c = 2 \times 0.1 \pi$$

$$\omega_c = 0.2\pi$$

putting ω_c in (1)

$$\Omega = \frac{2}{T} \tan \frac{0.2\pi}{2}$$

$$\Omega = \frac{2}{10^{-4}} \tan$$

$$\Omega = 61553.67 \text{ rad/sec}$$

$$\Omega_c = 6498.39 \text{ rad/sec}$$

If Sam freq given,
 then, $T = \frac{1}{F_s}$
 $= \frac{1}{10^4}$

Butterworth $\rightarrow$ All poles are filter.

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Butterworth :

1. $N = 2$
2. $\Omega_c = 6498.89$
3. Poles location

$$p_i = \pm \Omega_c e^{j(N+2i+1)\pi/2N}, \quad i = 0 \dots N-1$$

When $i = 0$

$$p_0 = \pm 6498.89 e^{j3\pi/4}$$

$$= \pm 6498.89 e^{j3\pi/4}$$

$$= \pm 6498.89 \left[\cos \frac{3\pi}{4} + j \sin \frac{3\pi}{4} \right]$$

$$p_0 = \pm 6498.89 [-0.707 + j0.707]$$

$$p_0^+ = -4595 + j4595$$

$$p_0^- = 4595 - j4595$$

When $i = 1$

$$p_1 = \pm \Omega_c e^{j(2+2+1)\pi/4}$$

$$p_1 = \pm 6498.89 e^{j5\pi/4}$$

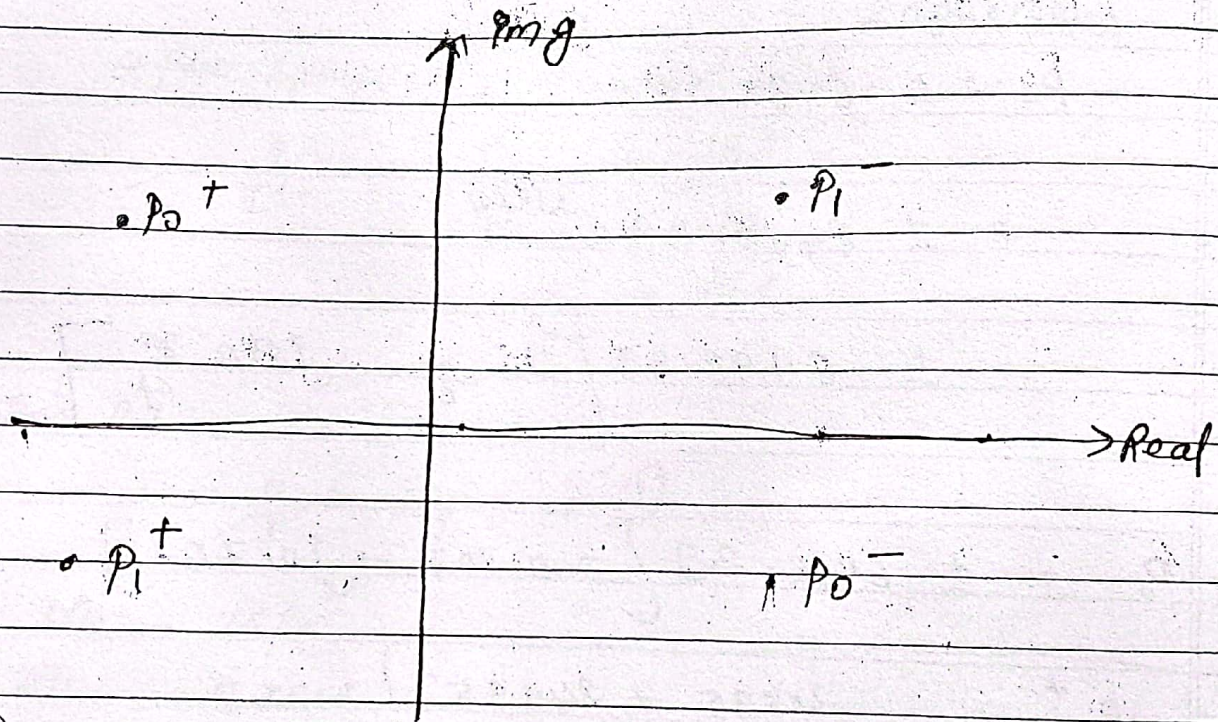
$$= \pm 6498.89 \left[\cos \frac{5\pi}{4} + j \sin \frac{5\pi}{4} \right]$$

$$= \pm 6498.89 [-0.707 - j0.707]$$

$$P_1^+ = -4595 - j4595$$

$$P_1^- = 4595 + j4595$$

Now plotting the poles:



Transfer function

$$H(s) = \frac{\Omega_c^N}{(s - P_0)(s - P_1)}$$

$$= \frac{(6498.39)^2}{(s + 4595 - j4595)(s + 4595 + j4595)}$$

$$= \frac{(6498.39)^2}{(s + 4595)^2 + (4595)^2}$$

$H(s) \xrightarrow{\text{BLT}} H(z)$

$$s \rightarrow \frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right]$$

$$\therefore H(z) = \frac{(6498.39)^2}{\left[\frac{2}{10^{-4}} \left[\frac{1-z^{-1}}{1+z^{-1}} \right] + 4895 \right]^2 + (4595)^2}$$

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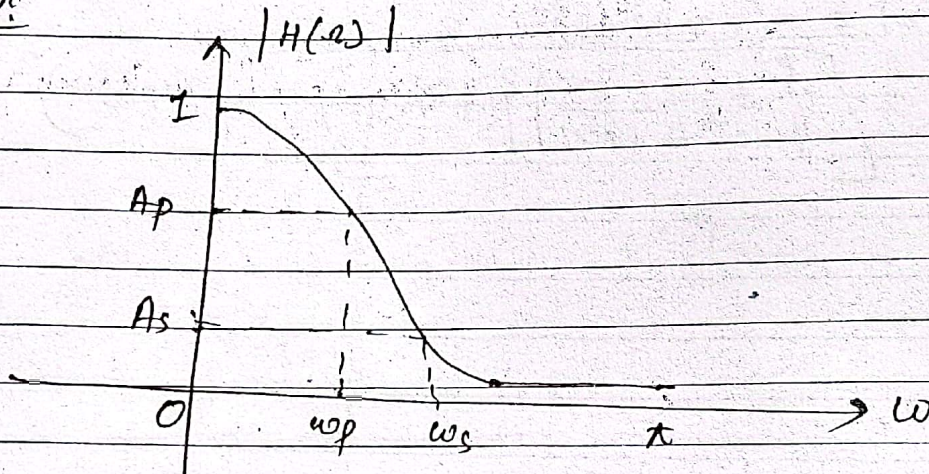
gnd.

Using BLT, design a digital filter using Butterworth.

$$0.8 \leq |H(e^{j\omega})| \leq 1, \text{ for } 0 \leq |\omega| \leq 0.2\pi$$

$$|H(e^{j\omega})| \leq 0.2, \text{ for } 0.6\pi \leq |\omega| \leq \pi$$

$\Rightarrow$ soln.



Given-

pass band attenuation (A_p) = 0.8

stop band attenuation (A_s) = 0.2

pass band frequency (ω_p) = 0.2π rad/sec

stop band frequency (ω_s) = 0.6π rad/sec

For BLT, we have,

$$\Omega = \frac{2}{T} \tan \frac{\omega}{2}$$

$$= \frac{2}{1} \tan \frac{\omega}{2} \quad [\because T = 1 \text{ sec}]$$

Since pass freq
& time is not
given

$$\Omega_p = 2 \tan \frac{0.2\pi}{2}$$

$$\Omega_p = 0.649 \text{ rad/sec}$$

$$\Omega_s = 2 \tan \frac{0.6\pi}{2}$$

$$\Omega_s = 2.75 \text{ rad/sec}$$

Butterworth Calculation

1. Order

$$N = \frac{\log \left[\frac{\frac{1}{A_s^2} - 1}{\frac{1}{A_p^2} - 1} \right]}{2 \log(\Omega_s / \Omega_p)}$$

$$= \frac{\log \left[\frac{\frac{1}{0.2^2} - 1}{\frac{1}{0.8^2} - 1} \right]}{2 \log(2.75 / 0.649)}$$

$$= \cancel{0.764}$$

$$N = 1.3$$

$$N \simeq 2$$

2. Cutoff frequency :

$$\Omega_c = \frac{\Omega_p}{\left[\frac{1}{A_p^2} - 1 \right]^{\frac{1}{2N}}}$$

$$= \frac{0.649}{\left[\frac{1}{0.82} - 1 \right]^{\frac{1}{2 \times 2}}}$$

$$\therefore \Omega_c = \cancel{0.85} \rightarrow 0.75$$

3. Poles location :

$$j(N+2i+1)\pi/2N \quad , i = 0, \dots, N-1$$

$$P_i = \pm \Omega_c e$$

when $i=0$

$$P_0 = \pm 0.75 e^{j3\pi/4}$$

$$= \pm 0.75 \left[\cos \frac{3\pi}{4} + j \sin \frac{3\pi}{4} \right]$$

$$= \pm 0.75 \left[-0.707 + j0.707 \right]$$

$$P_0^+ = -0.53 + j0.53$$

$$P_0^- = 0.53 - j0.53$$

When $p = 1$

$$p_1 = \pm 0.75 e^{j5\pi/4}$$

$$= \pm 0.75 \left[\cos \frac{5\pi}{4} + j \sin \frac{5\pi}{4} \right]$$

$$= \pm 0.75 [-0.707 - j 0.707]$$

$$p_1^+ = -0.53 - j0.53$$

$$p_1^- = 0.53 + j0.53$$

Since p_0^+ & p_1^+ lies in -ve side, we choose p_0^+ & p_1^+

$$H(s) = \frac{0.75^2}{(s+0.53-j0.53)(s+0.53+j0.53)}$$

$$= \frac{0.75^2}{(s+0.53)^2 + (0.53)^2}$$

$$H(s) \xrightarrow{\text{BLT}} H(z)$$

$$s \rightarrow \frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right]$$

$$H(z) = \frac{0.75^2}{\left(\frac{2(1-z^{-1})}{1+z^{-1}} + 0.53 \right)^2 + (0.53)^2}$$

$$\# \quad 0.89 < |H(e^{j\omega})| \leq 1, \quad 0 \leq N < 0.4\pi$$

$$|H(e^{j\omega})| \leq 0.18, \quad 0.6\pi \leq N \leq \pi$$

Use Impulse Invariance.

$\Rightarrow$ Soln:

Here; $A_p = 0.89$ $\omega_p = 0.4\pi$ rad/sample

$A_s = 0.18$ $\omega_s = 0.6\pi$ rad/sample

and

$$\Omega = \frac{\omega}{T}$$

$$, T=1$$

$$\Omega_p = 0.4\pi \text{ rad/sec}$$

$$\Omega_s = 0.6\pi \text{ rad/sec}$$

Now,

$$1. \quad N = \log \left[\frac{\frac{1}{A_s^2} - 1}{\frac{1}{A_p^2} - 1} \right]$$

$$2 \log (\Omega_s / \Omega_p)$$

$$N = 5.84$$

$$N \approx 6$$

2. Cutoff frequency

$$\Omega_c = \frac{\Omega_p}{\left[\frac{1}{A_p^2} - 1 \right]^{\frac{1}{2N}}}$$

$$= \frac{0.4}{\left[\frac{1}{0.89^2} - 1 \right]^{\frac{1}{2 \times 6}}}$$

$$\Omega_c = 1.404 \text{ rad/sec}$$

(Take angle from 0° to 90°).

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Since order $N=6$. So,

First pole, $\varphi = \frac{90^\circ}{6} = 15^\circ$, $\Delta = \frac{180^\circ}{6} = 30^\circ$

$\varphi_2 = 45^\circ$

$\varphi_3 = 75^\circ$

$\varphi_4 = 105^\circ$

$\varphi_5 = 135^\circ$

$\varphi_6 = 165^\circ$

$$H(s) = \frac{1}{(s^2 + 2\cos\varphi_1 s + 1)(s^2 + 2\cos\varphi_2 s + 1)(s^2 + 2\cos\varphi_3 s + 1)}$$

$$= \frac{1}{(s^2 + 2\cos 15^\circ s + 1)(s^2 + 2\cos 45^\circ s + 1)(s^2 + 2\cos 75^\circ s + 1)}$$

$$H(s) = \frac{1}{(s^2 + 1.931s + 1)(s^2 + 1.414s + 1)(s^2 + 0.517s + 1)}$$

$$H(s) = H(s) \Big|_{s \rightarrow \frac{s}{1.404}}$$

$$H(s) = \frac{1}{\left[\left(\frac{s}{1.404} \right)^2 + 1.931 \left(\frac{s}{1.404} \right) + 1 \right] \left[\left(\frac{s}{1.404} \right)^2 + 1.414 \left(\frac{s}{1.404} \right) + 1 \right] \left[\left(\frac{s}{1.404} \right)^2 + 0.517 \left(\frac{s}{1.404} \right) + 1 \right]}$$

1

$$\frac{1}{1.404} \left[s^2 + 1.981 \times 1.404s + 1.404^2 \right] \left[s^2 + 1.414 \times 1.404s + 1.404^2 \right]$$

$$\left[s^2 + 0.517 \times 1.404s + 1.404^2 \right]$$

$$= (1.404)^5$$

$$\left[s^2 + 1.981 \times 1.404s + 1.404^2 \right] \left[s^2 + 1.414 \times 1.404s + 1.404^2 \right] \left[s^2 + 0.517 \times 1.404s + 1.404^2 \right]$$

#

1.) Ripple Factor

$$\epsilon = \left[\frac{1}{A_p^2 - 1} \right]^{0.5}$$

$$\epsilon = \left[10^{AP/10} - 1 \right]^{0.5} \quad [dB]$$

2.) N (order)

$$|H(\omega)| = -20 \log(\epsilon) - 6(N-1) - 20N \log(\Omega_s)$$

3.) Poles location

$$S_i = x_i + jy_i$$

where $x_i = r \cos \theta_i$
 $y_i = r \sin \theta_i$

$$\theta_i = \frac{(N + 2i + 1)\pi}{2N}$$

σLR

$$\sigma = \Omega_p \frac{\beta^2 - 1}{2\beta} \quad R = \Omega_p \frac{\beta^2 + 1}{2\beta}$$

$$\beta = \left[\frac{\sqrt{1 + \epsilon^2} + 1}{\epsilon} \right]^{1/N}$$

$$H(s) = \frac{i}{(s-s_0)(s-s_1)(s-s_2) \dots}$$

$$\left. \begin{array}{l} i = b_0 \text{ if } N \text{ is odd} \\ \frac{b_0}{\sqrt{1+\epsilon^2}} \text{ if } N \text{ is even} \end{array} \right\}$$

Design a Chebyshev LPF of 1 rad/sec bandwidth and following specs.

1. Pass band Ripple = 2 dB
2. Cut off frequency = 1 rad/sec
3. Stop band attenuation of 20 dB or greater beyond 1.3 rad/sec.

⇒ Soln:

Given

$$A_p = 2 \text{ dB}$$

$$A_s = 20 \text{ dB}$$

$$\Omega_c = 1 \text{ rad/sec.}$$

$$\Omega_s = 1.3 \text{ rad/sec.}$$

Now

$$E = \left[10^{A_p/20} - 1 \right]^{0.5}$$

$$= \left[10^{2/20} - 1 \right]^{0.5}$$

$$E = 0.765$$

$$|H(\Omega)| = -20 \log_{10}(E) - 6(N-1) - 20N \log_{10}(\Omega_s)$$

$$-20 = -20 \log_{10}(0.765) - 6(N-1) - 20N \log_{10}(1.3)$$

$$N = 3.41$$

$$N \approx 4$$

$$S_i = x_i + jy_i$$

To find β ,

$$\beta = \left[\frac{\sqrt{1 + \epsilon^2} + 1}{\epsilon} \right]^{1/N}$$

$$\beta = 1.31$$

and

$$\begin{aligned} \sigma &= \frac{\Omega_p \beta^2 - 1}{2\beta} \\ &= \frac{1 \cdot 1.31^2 - 1}{2 \times 1.31} \end{aligned}$$

$$\sigma = 0.273$$

for

$$\begin{aligned} R &= \frac{\Omega_p \beta^2 + 1}{2\beta} \\ &= \frac{1 \cdot 1.31^2 + 1}{2 \times 1.31} \end{aligned}$$

$$R = 1.04$$

also,

$$\theta_i = \frac{(N + 2i + 1)\pi}{2N}$$

$$i = 0, 1, 2, 3$$

$$\theta_0 = \frac{5\pi}{8}$$

$$\theta_1 = \frac{7\pi}{8}$$

$$\theta_2 = \frac{9\pi}{8}$$

$$\theta_3 = \frac{11\pi}{8}$$

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$$x_i = r \cos \theta_i$$

$$x_0 = 0.273 \cos \frac{5\pi}{8} = -0.10$$

$$x_1 = 0.273 \cos \frac{7\pi}{8} = -0.28$$

$$x_2 = 0.273 \cos \frac{9\pi}{8} = -0.25$$

$$x_3 = 0.273 \cos \frac{11\pi}{8} = -0.10$$

and

$$y_i = r \sin \theta_i$$

$$y_0 = 1.004 \sin \frac{5\pi}{8} = 0.96$$

$$y_1 = 0.4$$

$$y_2 = -0.4$$

$$y_3 = -0.96$$

Now finally

$$s_i = x_i + jy_i$$

$$s_0 = -0.10 + j0.96$$

$$s_1 = -0.25 + j0.4$$

$$s_2 = -0.25 - j0.4$$

$$s_3 = -0.10 - j0.96$$

$$H(s) = \frac{1}{(s+0.10 - j0.96)(s+0.25 - j0.4)(s+0.25 + j0.4)(s+0.10 + j0.96)}$$

Since $N = \text{even}$ so formula for P is,

$$P = \frac{b_0}{\sqrt{1+e^2}}$$

[and to find b_0 , multiply all denominator in above $H(s)$ equation, then the constant number from the result is b_0 .]

IMP

#

The transfer function of analog LPF is,
 $H_a(s) = \frac{1}{s+1}$ & its b/w is 1 rad/sec.

Design a IIR filter using BLT method whose cutoff frequency is 0.2π and sampling interval is 0.0167 sec, also consider warping effect.

⇒ Soln:

$$H(s) \xrightarrow{\text{BLT}} H(z)$$

$$s \rightarrow \frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right]$$

$$\text{digital freq } (\omega_c) = 0.2\pi \text{ rad/sec.}$$

$$\Omega_c = ?$$

$$\Omega_c = \frac{2}{T} \tan \frac{\omega_c}{2}$$

$$= \frac{2}{0.0167} \tan \frac{0.2\pi}{2}$$

$$\Omega_c = 38.91 \text{ rad/sec.}$$

So, transforming the cutoff frequency of $H_a(s)$ into
 $\Omega_c = 38.91$

$$H(s) = H_a(s) \Big|_{s \rightarrow \frac{s}{38.91}}$$

$$= \frac{1}{\frac{s}{38.91} + 1}$$

$$H(s) = \frac{38.91}{s + 38.91}$$

So using BLT,

$$H(z) = H(s) \Big|_{s \rightarrow \frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right]}$$

$$= \frac{38.91}{\frac{2}{0.0167} \left[\frac{1-z^{-1}}{1+z^{-1}} \right] + 38.91}$$

$$= \frac{38.91}{119.76 \left[\frac{1-z^{-1}}{1+z^{-1}} \right] + 38.91}$$

$$H(z) = \frac{38.91}{119.76 \left[\frac{1-z^{-1}}{1+z^{-1}} \right] + 38.91}$$

$$= \frac{38.91}{119.76 \left[\frac{1-z^{-1}}{1+z^{-1}} \right] + 38.91}$$

$H(s) = \frac{s+0.1}{(s+0.1)^2 + 16}$, obtain digital IIR

filter using BLT, which is resonant at
 $\omega_r = \frac{\pi}{2}$

⇒ Soln:

let find the resonance freq. of $H(s)$,

$$(s+0.1)^2 + 16 = 0$$

$$\text{or, } (s+0.1)^2 = -16$$

$$\text{or, } s+0.1 = \pm 4j$$

$$\text{or, } s = -0.1 \pm 4j$$

So its resonance frequency is 4 rad/sec

$$\Omega_r = \frac{2}{T} \tan \frac{\omega_r}{2}$$

$$\text{or, } 4 = \frac{2}{T} \tan \frac{\pi/2}{2}$$

$$\therefore T = 0.5 \text{ sec.}$$

Using BLT,

$$H(z) \xrightarrow{\quad}$$

$$H(z) = H(s) \Big|_{s \rightarrow \frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right]}$$

$$H(z) = \frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right] + 0.1$$

$$\left\{ \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right\}^2 + 16$$

$$H(z) = \frac{\frac{2}{0.5} \left[\frac{1 - z^{-1}}{1 + z^{-1}} \right] + 0.1}{\left\{ \frac{2}{0.5} \left(\frac{1 - z^{-1}}{1 + z^{-1}} \right) \right\}^2 + 16}$$

Digital Signal Analysis and Processing (3-1-2)

Evaluation:

Theory	Practical	Total
Sessional	30	20
Final	50	50
Total	80	20
		100

Course Objectives:

1. To provide knowledge on digital signal processing techniques.
2. To design and implement IIR and FIR digital filter.

Course Contents:

1. **Signals, Systems and Signal Processing** (5 hrs)
 - 1.1 Basic elements of Digital Signal Processing
 - 1.2 Energy signal, Power signal
 - 1.2 Need of Digital Signal Processing over Analog Signal Processing
 - 1.4 Sampling continuous signals and spectral properties of sampled signals.
2. **Discrete-time Signals and System** (9 hrs) 49
 - 2.1 Elementary discrete-time signals
 - 2.2 Discrete time Fourier series and properties
 - 2.3 Discrete time Fourier transform and properties
 - 2.4 Discrete time system properties
 - 2.5 Properties of Linear Time-Invariant systems (LTI)
 - 2.6 LTI convolution sum characterized by constant coefficient difference equations
 - 2.7 Stability of LTI systems, Implementation of LTI system.
 - 2.8 Frequency Response of LTI systems
3. **Review of Z-Transform** (4 hrs) 115
 - 3.1 Definition z-transform
 - 3.2 Convergence of Z-transform, Region of convergence
 - 3.3 Properties of Z-Transform (linearity, time shift, multiplication by exponential sequence, differentiation, time reversal, convolution, multiplication)
 - 3.4 Inverse z-transform-by long division, by partial fraction expansion
4. **Discrete Fourier Transform** (8 hrs) 153
 - 4.1 Definition and application
 - 4.2 Frequency response of LTI system
 - 4.3 Forward and Reverse transform
 - 4.4 Properties of the Discrete Fourier Transform: linearity and Symmetry, time shift, frequency shift, duality, convolution, multiplication, conjugation & conjugate symmetry
 - 4.5 Basic concept of Fast Fourier Transform (FFT) algorithm

(6 hrs)

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5. Discrete Filter Structure

- 5.1 FIR filter overview
- 5.2 Structures for FIR Filter (direct, cascade, frequency sampling, lattice)
- 5.3 IIR Filter overview
- 5.4 Structure for IIR filter (direct form I & II, cascade, lattice, lattice ladder)
- 5.5 Quantization of filter coefficients and effects on location of poles, and zeros

(7 hrs)

257
6. FIR Filter Design

- 6.1 Gibbs phenomena in FIR filter design
- 6.2 Filter Design by Window method (rectangular window, hanning window, hamming window)
- 6.3 Filter design by Kaiser window
- 6.4 Filter design by frequency sampling
- 6.5 Filter design using the Remez exchange algorithm

273
7. IIR Filter Design

- 7.1 Filter design using low pass approximations Butterworth filter.
- 7.2 Filter design using impulse invariance method
- 7.3 Filter design using bilinear transformation
- 7.2 Properties of Chebyshev & Elliptic filters
- 7.3 High pass, Band pass and Notch filters.

(6 hrs)

Laboratory

- 1. Overview of DSP tools
- 2. Scaling, dynamic range and noise behavior of a recursive digital filter
- 3. Response of a non-recursive digital filter, Implementation in Impulse Invariant and Bil Transformation.
- 4. Bandpass filters implemented using cascade second order systems
- 5. Design of FIR filter
- 6. Design of IIR filter

Reference Books:

- 1. A.V. Oppenheim, *Discrete-Time Signal Processing*, Prentice Hall, 1990.
- 2. J.G. Proakis and D.G. Manolakis, *Digital Signal Processing*, Prentice Hall of India.
- 3. S.K. Mitra, *Digital Signal Processing, A Computer-based Approach*, McGraw Hill.